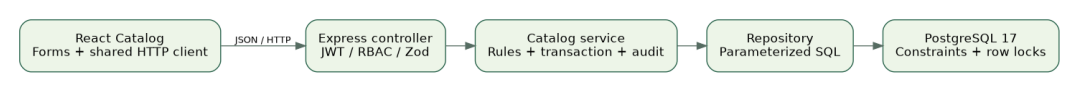
# BookFlow: Catalog and API Contract

- Phạm Văn Tú · 52400252
- REST / GraphQL / SOAP comparison
- A connected Catalog with safe concurrent edits

# Model the physical library, not just a list

- Category classifies a title
- Book stores metadata; Copy stores a unique barcode
- Archiving preserves history



# REST, GraphQL and SOAP: compare the same task

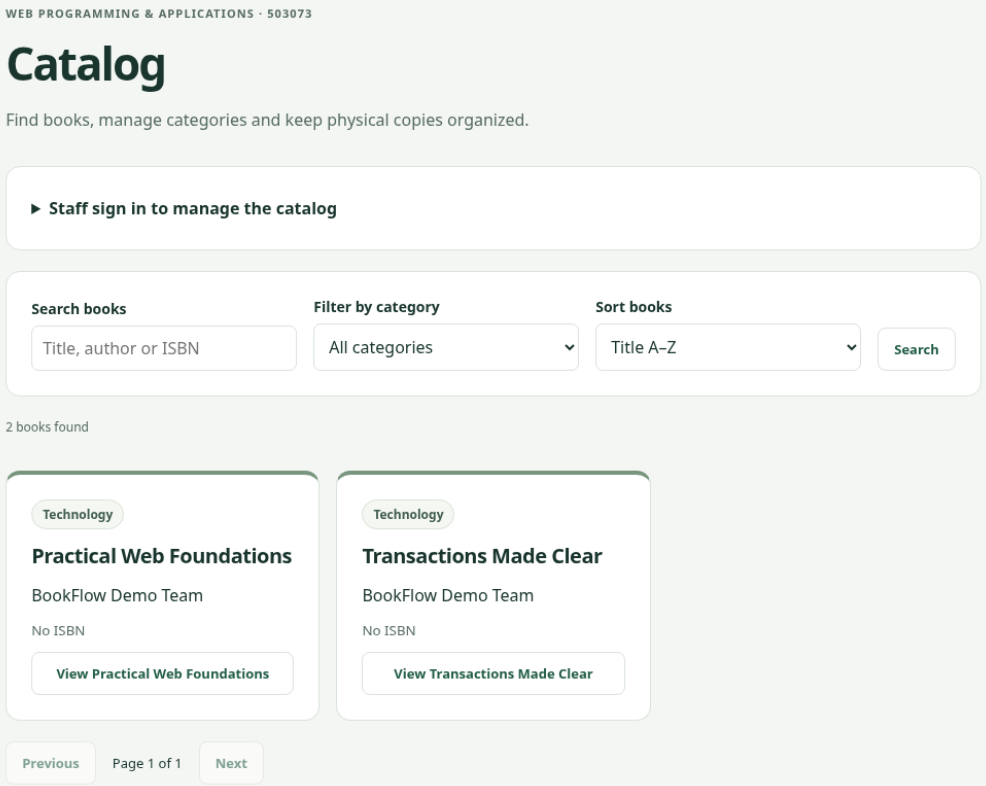
- REST: resources, methods and HTTP validators
- GraphQL: typed field selection; resolver and query limits
- SOAP: XML messaging and operation contracts
- No performance ranking without an equivalent benchmark

# A contract the browser can use

- GET reads; POST creates; PUT replaces; PATCH changes a subset
- 201 + Location for creation; 204 without a response body
- 400 invalid · 403 forbidden · 409 conflict
- OpenAPI + Postman describe implemented operations

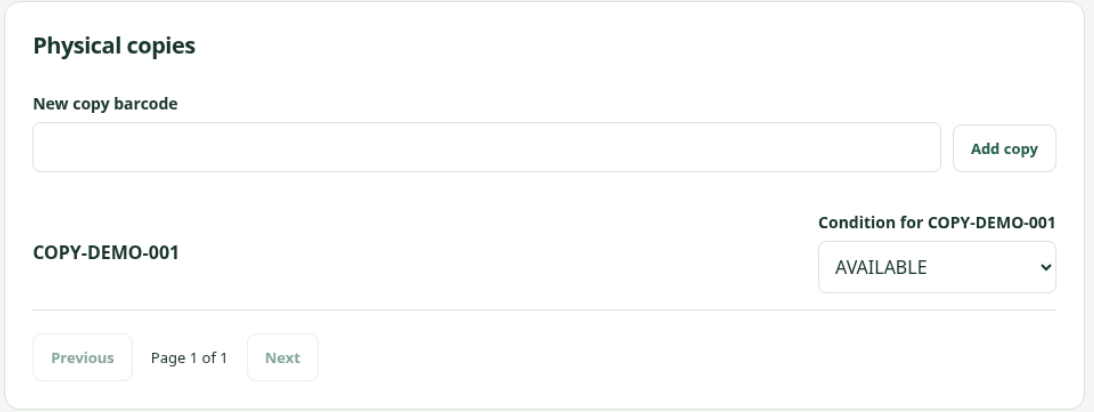
# Public UI and staff access

- Search title / author / ISBN
- Filter, allowlisted sort and bounded pagination
- Public reads; LIBRARIAN/ADMIN mutations
- Server permissions are authoritative



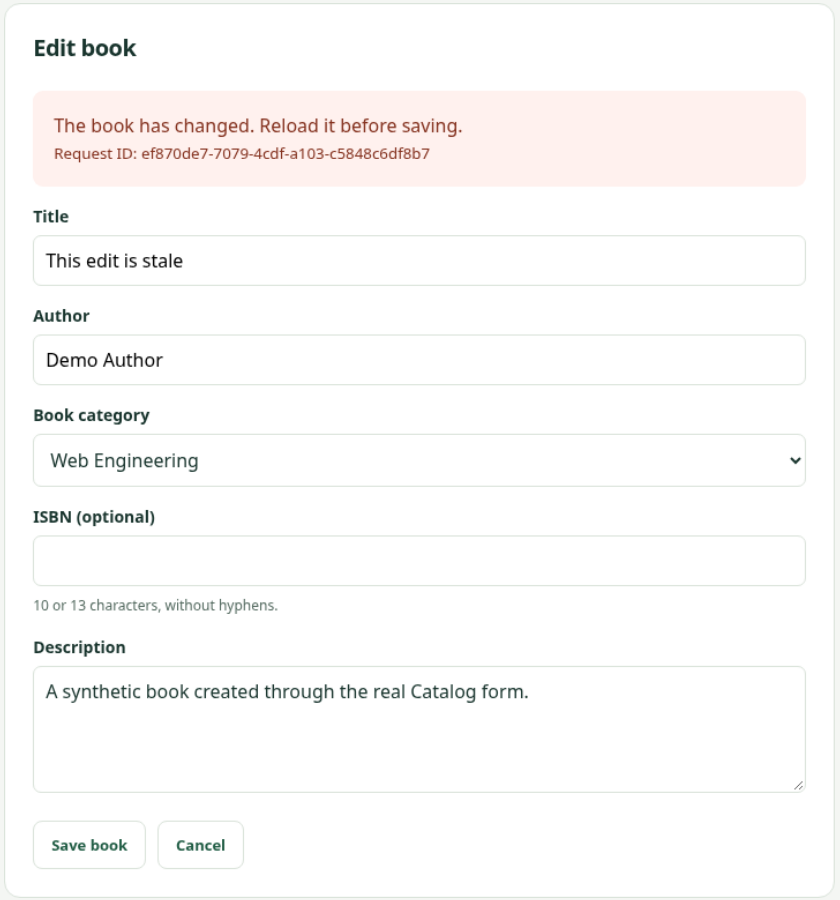
# Demonstrate a real management flow

- Create a category and book
- Add a physical copy; change condition
- Availability changes independently of metadata
- Duplicate barcode returns 409



# Two editors: detect a stale version

- GET returns a strong metadata ETag
- Mutation sends If-Match
- One writer advances v1 → v2
- A stale writer receives 412 and reloads



# Protect history and open loans

- Archive a title; do not erase its history
- Referenced categories cannot be deleted
- Open loans block archive and copy-condition changes
- Audit commits with the mutation

# Evidence: separate local checks from CI

- 59 foundation tests + 52 integration + 10 browser = 121 passed
- 24 Catalog integration cases
- Desktop/mobile flows, errors and conflict recovery
- No invented performance or production claims

# Integration and remaining group work

- Catalog is implemented; loan flows remain separate
- Refresh/logout and full account UI remain planned
- Combine chapters and apply the Faculty template
- Record all three members in English; total ≤20 minutes